Worksheet 3C: Quadratics — Real-World Applications

What This Worksheet Is About

Quadratics aren't just abstract algebra — they show up everywhere: - Objects flying through the air follow parabolic paths - Trigonometric functions can be approximated by polynomials - Probability distributions use binomial expansions - Compound growth involves expanding expressions like $(1 + r)^n$

This worksheet connects the algebra you've learned to real applications.

Part A: Projectile Motion (12 problems)

When you throw a ball, drop an object, or launch a rocket, its height follows a quadratic equation:

$$h(t) = h_0 + v_0 t - \frac{1}{2} g t^2$$

Where: - $h(t)$ = height at time t - h_0 = initial height (starting height) - v_0 = initial velocity (positive = upward, negative = downward) - g = acceleration due to gravity (≈ 10 m/s^2 or 32 ft/s^2) - t = time

Simplified form (using $g = 10$ m/s²):

$$h(t) = h_0 + v_0 t - 5t^2$$

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1. A ball is thrown upward from ground level with initial velocity 20 m/s.

$$h(t) = 0 + 20t - 5t^2 = 20t - 5t^2$$

- (a) What is the height at $t = 1$ second?
- (b) What is the height at $t = 2$ seconds?
- (c) What is the height at $t = 4$ seconds? What does this answer tell you?

2. Using the same equation $h(t) = 20t - 5t^2$:

- (a) Factor the right side: $h(t) =$ _____
- (b) Set $h(t) = 0$ and solve. What do the two solutions represent physically?

3. A ball is dropped from a 45-meter building (no initial velocity).

$$h(t) = 45 - 5t^2$$

- (a) How high is the ball after 1 second?
- (b) How high after 2 seconds?
- (c) When does the ball hit the ground? (Set $h(t) = 0$ and solve)

4. A ball is thrown downward from a 80-meter cliff with initial velocity 10 m/s downward.

$$h(t) = 80 - 10t - 5t^2$$

- (a) Why is the v_0 term negative?
- (b) When does the ball hit the ground?
- (c) How fast is it falling? (This requires calculus, but estimate: is it speeding up or slowing down?)

5. A rocket is launched upward with initial velocity 50 m/s from a platform 20 meters high.

$$h(t) = 20 + 50t - 5t^2$$

- (a) Rewrite in standard form: $h(t) = -5t^2 + 50t + 20$
- (b) What is $h(0)$? Does this match the problem description?
- (c) Use the quadratic formula to find when $h(t) = 0$.
- (d) Why do you get two answers? Which one makes physical sense?

6. Finding Maximum Height

For $h(t) = -5t^2 + 20t$, the maximum occurs at $t = \frac{-b}{2a}$.

- (a) Identify a and b from the equation.
- (b) Calculate $t_{max} = \frac{-b}{2a}$
- (c) Find the maximum height by plugging t_{max} back into $h(t)$.

7. For the rocket in problem 5, $h(t) = -5t^2 + 50t + 20$:

- (a) At what time does it reach maximum height?
- (b) What is the maximum height?

8. A football is kicked with the path $h(t) = -5t^2 + 15t + 1$ (height in meters).

- (a) What was the initial height? (The kicker's foot height)
- (b) When does it reach maximum height?
- (c) What is the maximum height?
- (d) When does it hit the ground?

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9. Horizontal Distance

When launched at an angle, a projectile's horizontal distance is:

$$x(t) = v_x \cdot t$$

And vertical height is:

$$y(t) = v_y \cdot t - 5t^2$$

A ball is kicked with $v_x = 10$ m/s (horizontal) and $v_y = 15$ m/s (vertical).

- (a) Write the equation for $y(t)$.
- (b) When does the ball land ($y = 0$)?
- (c) How far horizontally has it traveled when it lands?

10. Eliminate t from the equations in problem 9:

- From $x = 10t$, we get $t = \frac{x}{10}$
- Substitute into $y(t)$: $y = 15 \cdot \frac{x}{10} - 5 \cdot \left(\frac{x}{10}\right)^2$
- Simplify to show that y is a quadratic function of x (a parabola!)

11. A ball thrown from height 2m follows $h(t) = 2 + 12t - 5t^2$.

Write an equation to answer: "When is the ball at height 10m?" Then solve it.

12. Two balls are thrown: - Ball A: $h_A(t) = 30t - 5t^2$ (from ground, fast) - Ball B: $h_B(t) = 25 + 10t - 5t^2$ (from 25m platform, slower)

When are they at the same height? (Set $h_A = h_B$ and solve)

Part B: Area Problems (8 problems)

13. A rectangular garden has length 3 meters more than its width. Let w = width.

- (a) Write an expression for the length in terms of w.
- (b) Write an expression for the area A in terms of w.
- (c) If the area is 70 m², write and solve an equation to find the dimensions.

14. A farmer has 100 meters of fencing to enclose a rectangular field against a river (no fence needed on the river side).

- (a) If the width is w, what is the length in terms of w? (Hint: $2w + L = 100$)
- (b) Write the area as a function of w.
- (c) This is a quadratic! Find the width that maximizes the area.
- (d) What is the maximum area?

15. A square has its side length increased by 4 cm. The new area is 81 cm².

- (a) Let s = original side length. Write an equation.
- (b) Solve for s.

16. The product of two consecutive positive integers is 182. Find them.

(Let n = first integer, so $n+1$ = second integer)

17. The product of two consecutive even integers is 288. Find them.

18. A rectangular photo is 4 cm longer than it is wide. A frame of uniform width 2 cm surrounds it. The total area (photo + frame) is 165 cm².

- (a) If the photo width is w, what are the outer dimensions?
- (b) Write an equation for the total area.
- (c) Solve for the photo dimensions.

19. A box with no lid is made from a 20 cm × 30 cm sheet by cutting squares of side x from each corner and folding up.

- (a) What are the dimensions of the box in terms of x?
- (b) Write the volume as a function of x.
- (c) If the volume must be 1008 cm³, find x.

20. The sum of a number and its square is 72. Find all possible values.

Part C: Polynomial Approximations of Trig Functions (10 problems)

Key Insight: Sine, cosine, and other transcendental functions can be approximated by polynomials! These are called **Taylor series** approximations.

For small values of θ (in radians):

$$\sin(\theta) \approx \theta - \frac{\theta^3}{6} + \frac{\theta^5}{120} - \cdots$$

$$\cos(\theta) \approx 1 - \frac{\theta^2}{2} + \frac{\theta^4}{24} - \cdots$$

These are just polynomials! All your algebra skills apply.

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21. Using the first two terms of the sine approximation: $\sin(\theta) \approx \theta - \frac{\theta^3}{6}$

- (a) Calculate this approximation for $\theta = 0.1$ radians.
- (b) Your calculator says $\sin(0.1) = 0.0998334...$. How close is your approximation?

22. Using the first two terms of the cosine approximation: $\cos(\theta) \approx 1 - \frac{\theta^2}{2}$

- (a) Calculate this approximation for $\theta = 0.2$ radians.
- (b) Compare to the actual value $\cos(0.2) = 0.980067...$

23. Factor θ out of the sine approximation $\theta - \frac{\theta^3}{6}$:

$$\sin(\theta) \approx \theta\left(1 - \frac{\theta^2}{6}\right)$$

What does this tell you about $\sin(\theta)$ when θ is very small?

24. The cosine approximation $1 - \frac{\theta^2}{2}$ is a quadratic in θ.

- (a) What is the "vertex" of this parabola?
- (b) Does this match what you know about the cosine function at $\theta = 0$?

25. For small angles, engineers often use $\sin(\theta) \approx \theta$ and $\cos(\theta) \approx 1$.

- (a) For what range of θ is this reasonable? (When does $\frac{\theta^2}{2}$ become significant?)
- (b) If you need $\cos(\theta)$ to be accurate within 1%, how small must θ be?

26. The tangent function can be approximated as:

$$\tan(\theta) \approx \theta + \frac{\theta^3}{3}$$

- (a) Factor out θ.
- (b) Calculate $\tan(0.1)$ using this approximation.
- (c) Compare to actual: $\tan(0.1) = 0.1003347...$

27. Combining Approximations

Using $\sin(\theta) \approx \theta - \frac{\theta^3}{6}$ and $\cos(\theta) \approx 1 - \frac{\theta^2}{2}$:

Calculate $\sin^2(\theta) + \cos^2(\theta)$ for $\theta = 0.1$ using the approximations.

Does it equal 1? (It should be close!)

28. The approximation $e^x \approx 1 + x + \frac{x^2}{2}$ is also a polynomial.

- (a) Calculate $e^{0.1}$ using this approximation.
- (b) Actual value: $e^{0.1} = 1.10517...$. How close?

29. Using $e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$:

- (a) This is a polynomial of degree _____.
- (b) Calculate $e^{0.5}$ using all four terms.
- (c) Actual value: $e^{0.5} = 1.6487...$. How's the accuracy?

30. Connection to Euler's Formula

The famous formula $e^{i\theta} = \cos(\theta) + i\sin(\theta)$ connects exponentials and trig.

Using the polynomial approximations, verify this is consistent for small θ by comparing: - $e^{i\theta} \approx 1 + i\theta + \frac{(i\theta)^2}{2} = 1 + i\theta - \frac{\theta^2}{2}$ (since $i^2 = -1$) - $\cos(\theta) + i\sin(\theta) \approx \left(1 - \frac{\theta^2}{2}\right) + i(\theta)$

Are they the same?

Part D: Binomial Expansion and Probability (10 problems)

Connection: When you flip a coin n times, the probability of getting exactly k heads involves the binomial coefficient from Pascal's Triangle!

$$P(k \text{ heads in } n \text{ flips}) = \binom{n}{k} p^k (1-p)^{n-k}$$

Where $\binom{n}{k}$ is the entry in row n, position k of Pascal's Triangle.

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31. Expand $(H+T)^3$ where H and T represent heads and tails.

This represents all possible outcomes of 3 coin flips!

32. From your expansion, identify: - (a) How many ways to get 0 heads (all tails)? - (b) How many ways to get exactly 1 head? - (c) How many ways to get exactly 2 heads? - (d) How many ways to get 3 heads? - (e) Total number of outcomes?

33. For a fair coin ($p = 0.5$), expand $(0.5+0.5)^3$ to find the probability of each outcome.

34. A biased coin has $P(\text{heads}) = 0.6$ and $P(\text{tails}) = 0.4$.

Expand $(0.6+0.4)^3$ to find: - (a) $P(0 \text{ heads})$ - (b) $P(1 \text{ head})$ - (c) $P(2 \text{ heads})$ - (d) $P(3 \text{ heads})$

35. Using Pascal's Triangle row 4: **1, 4, 6, 4, 1**

For 4 flips of a fair coin, calculate: - (a) $P(\text{exactly 2 heads})$ - (b) $P(\text{at least 3 heads})$

36. Expand $(p+q)^4$ where $p+q=1$.

This is the general probability distribution for 4 trials!

37. A multiple choice test has 5 questions, each with 4 options (so $P(\text{correct by guessing}) = 0.25$).

Using the binomial expansion, what's the probability of getting exactly 3 correct by pure guessing?

(Use row 5 of Pascal's Triangle: 1, 5, 10, 10, 5, 1)

38. In the same test, what's the probability of getting at least 4 correct by guessing?

39. Expected Value Connection

For n flips of a fair coin, the expected number of heads is $\frac{n}{2}$.

Verify this makes sense: In $(H+T)^4$ expanded, which term has the largest coefficient? What does this represent?

40. A basketball player makes 70% of free throws. Using $(0.7+0.3)^3$: - (a) What's the probability of making all 3 shots? - (b) What's the probability of making at least 2?

Part E: Compound Growth and Finance (10 problems)

The Compound Interest Formula:

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

Where: - A = final amount - P = principal (initial amount) - r = annual interest rate (as decimal) - n = number of times compounded per year - t = time in years

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41. You invest €1000 at 5% annual interest, compounded yearly, for 2 years.

$$A = 1000(1 + 0.05)^2 = 1000(1.05)^2$$

- (a) Expand $(1.05)^2$ using $(1 + 0.05)^2 = 1 + 2(0.05) + (0.05)^2$
- (b) Calculate the final amount.
- (c) How much was earned in interest?

42. Same investment, but for 3 years: $A = 1000(1.05)^3$

- (a) Expand $(1.05)^3$ using Pascal's Triangle coefficients: 1, 3, 3, 1
- (b) Calculate the final amount.

43. The Rule of 72

To estimate how long it takes money to double at interest rate $r\%$:

$$\text{Doubling time} \approx \frac{72}{r}$$

- (a) At 6% interest, approximately how many years to double?
- (b) Verify: Calculate $(1.06)^{12}$ (you can use the approximation or calculator).

44. Expand $(1 + r)^2$ symbolically.

If $r = 0.1$ (10% return), calculate the growth factor after 2 years.

45. For small r, the approximation $(1 + r)^n \approx 1 + nr$ is sometimes used.

- (a) Using this approximation, what is $(1.02)^5$?
- (b) Using exact expansion (or calculator), what is $(1.02)^5$?
- (c) When is the approximation good enough?

46. Inflation and Purchasing Power

If inflation is 3% per year, the purchasing power of €100 after n years is:

$$\text{Value} = 100 \cdot (1 - 0.03)^n = 100 \cdot (0.97)^n$$

- (a) What is the purchasing power after 2 years?
- (b) After 5 years?
- (c) Approximately how many years until the value is halved?

47. Population Growth

A town's population grows 2% per year from an initial 50,000.

$$P(t) = 50000(1.02)^t$$

- (a) What's the population after 3 years?
- (b) Using logarithms or trial, when does it reach 60,000?

48. Depreciation

A car worth €20,000 depreciates 15% per year.

$$V(t) = 20000(1 - 0.15)^t = 20000(0.85)^t$$

- (a) Value after 1 year?
- (b) Value after 3 years?
- (c) When is it worth less than €10,000?

49. Continuous Compounding Limit

As $n \to \infty$, $\left(1 + \frac{1}{n}\right)^n \to e \approx 2.718$

Calculate for: - (a) $n = 1$ - (b) $n = 2$ - (c) $n = 10$ - (d) $n = 100$

Watch it approach e!

50. Loan Payments

A simplified loan payment formula involves:

$$\text{Payment} = P \cdot \frac{r(1+r)^n}{(1+r)^n - 1}$$

For a €10,000 loan at 0.5% monthly interest ($r = 0.005$) for 24 months:

- (a) Calculate $(1.005)^{24}$ (use calculator or approximation)
- (b) Find the monthly payment.
- (c) What's the total paid over 24 months?
- (d) How much was interest?
